

DANIEL GUO

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Education

University of California, Los Angeles (UCLA) | *Computer Science* **Sep. 2023 – Expected June 2026**

- Data Structures and Algorithms, Computer Organization, Software Construction, Operating Systems, Computer Graphics, Computer Animation, Data Science, Neural Networks and Deep Learning
- Dean's Honor List 2024

Experience

Trustworthy AI Lab | *Student Researcher* **March 2024 – Present**

- Performed runtime and memory analysis; enhanced data loading efficiency of CTGAN (conditional GAN) by 100 times through with **Dask** Delayed objects while ensuring compatibility with **PyTorch** tensor datasets
- Utilized **PyTest** to rigorously test column transformers and reverse transformations, ensuring model integrity and reliability throughout pipeline, deploying Python scripts with **Docker** to guarantee reliable operation

Upsilon Pi Epsilon (UPE) | *Web Development Officer* **January 2025 – Present**

- Designed and implemented the website using a full-stack approach, leveraging **Python** and the **Django** framework for backend logic and database management, **JavaScript** for interactive frontend elements, and **HTML/CSS** for structuring and styling the user interface for CS students with over 2000 website views
- Developed and maintained the test bank system, ensuring secure storage and access to academic resources for UPE members; implemented **Linux DevOps** practices for server management, deployment, and maintenance

Cyber Summer Research Fellowship | *Cybersecurity Researcher* **July 2024 – September 2024**

- Developed convolutional neural network using **PyTorch** to detect image steganography (LSB, PVD, DCT, FFT), applying transfer learning through established models (EfficientNet), achieving over 87% accuracy in multiclass classification on over 33,000 image dataset
- Co-authored Deep Learning for Steganographic Image Detection, accepted by 2024 IEEE MIT URTC Conference

Purrlicious Boba | *Full-stack Developer – Freelance* **June 2024 – Present**

- Designed and developed frontend for a local cat cafe and boba shop, using **React** with **TypeScript**, **HTML/CSS**, incorporating type safety, maintainability, and responsive design
- Built and deployed the backend infrastructure on **AWS** cloud services for scalability, and integrated the **Gmail API** for automated newsletter distribution and **MongoDB** for data management of shop information and drink menu

Glocol Networks | *Software Intern* **June 2022 – July 2022**

- Engineered real-time occupancy detection for system mass transit vehicles using **Python** algorithms, Bluetooth signals with Raspberry Pi devices to collect beacon data from mobile devices, deployed on **AWS** IoT Greengrass Lambda
- Extracted Amtrak train departure and arrival times and displayed on frontend with **JavaScript**, **Vue.js**, **HTML/CSS**

Projects

UCLA ACM AI | *Python, PyTorch, Pandas, Pillow, NumPy* **January 2024 - Present**

- Led a 3-member team, leveraging Python, PyTorch, and NumPy, to design and implement a Wasserstein Generative Adversarial Network (WGAN) model for generating and discriminating emojis from dataset of over 10,000 images
- Implemented GPT-2 from scratch with PyTorch, including embeddings, attention mechanisms, and stochastic decoding
- Oversaw the training and evaluation of both the WGAN and GPT-2 models, including hyperparameter tuning, performance analysis, and model selection

Cup Phong | *Javascript, THREE.js*

- Developed interactive 3D game environment alongside **Javascript** and **THREE.js**, implementing features such as camera movement, collision detection, particle systems, and real-time physics for a dynamic gaming experience

BruinBowl | *MongoDB, Express, React, Node* **March 2024 - June 2024**

- Collaborated with 6 students to design interactive web application on UCLA trivia, using Git for version control
- Designed user and gameplay functionality using **MongoDB**, **Express**, **React**, **Node**; created secure user login and authentication alongside features scalable for hundreds of users: question search, real-time leaderboards, and user profiles

Technical Skills

Languages: Python, C/C++, JavaScript, HTML/CSS, Java, SQL, Bash, Verilog

Developer Tools: Git, Docker

Technologies/Frameworks: Vue.js, MongoDB, Express.js, React, Node.js, PyTorch, NumPy, Pandas, Dask, Tailwind CSS, AWS